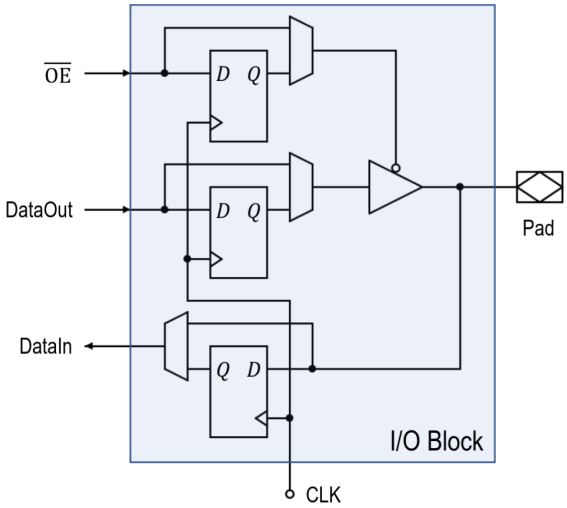
1 Lesson 16-10-26

1.1 Complex Programmable Logic Devices

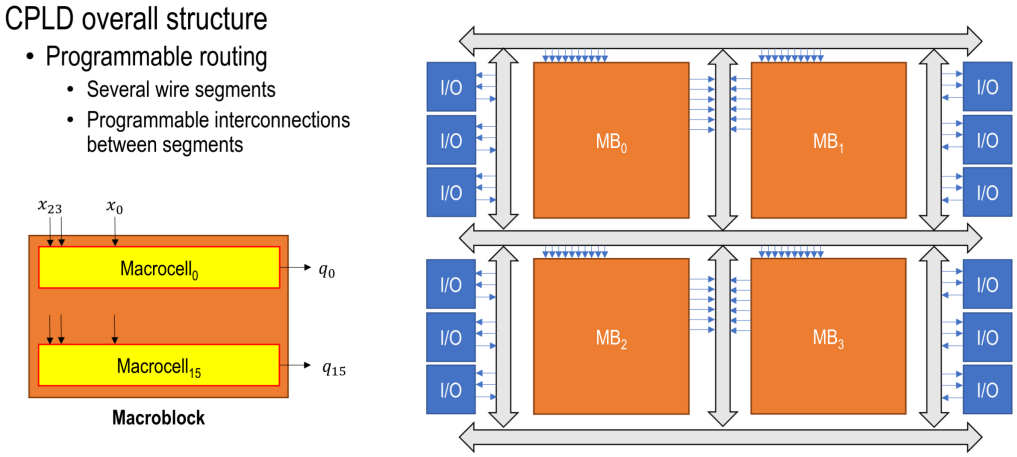
In complex circuits, the number of inputs and outputs increases. Arithmetic operations usually require many signals, such that delays increase as the number of inputs grows. Noting that functions operate on a subset of the total number of inputs, we see that it is not necessary to have a big PAL (or PLA) with all functions implement.

We said that grouping together multiple macrocells formed a logical unit known as a macroblock. Now, by grouping together multiple macroblocks, we get a *Complex Programmable Logic Devices* (**CPLD**). CPLDs need programmable interconnections to route signals from one Macrocell to another, ithin the same logic unit or between different logic units.

First off, we briefly talk about **I/O blocks**. When packaging logic devices into IC packages, for each pin we might like to have some circuitry that governs wether that pin is inputting or outputting data, and also acts as a synchronized buffer for that data. Such circuit is called an I/O block and shown below.



At this point we can package the actual macroblocks (shown left) toghether, interfacing them with the I/O blocks.



1.1.1 CPLD performance

The main problems of CPLDs is the restricted number of flip-flops: we cannot implement complex finite state machines, so we cannot implement complex sequential data paths, and naturally we cannot implement a processor (which requires lots of Flip Flops for the FSM in the control unit, and lots of Flip Flops for registers in the data path). As such, CPLDs used for glue logic, configured at low granularity.

1.2 Field Programmable Gate Arrays

We now get to **FPGA**s (*Field Programmable Gate Arrays*). Many digital circuits need large storage for temporary data: for instance pipeline registers, shift registers, counters, FSM registers, memories, FIFOs, register files, ... CPLDs do not provide enough storage

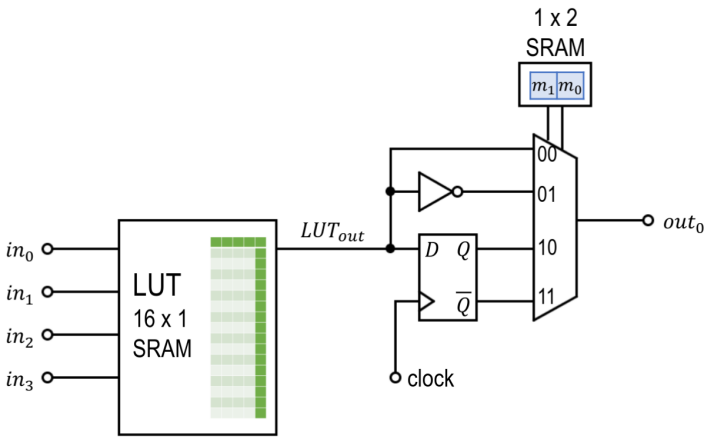
FPGAs are programmable logic devices based on *Look-Up Tables* (**LUT**s). A LUT is a small memory used to implement combinational logic functions, from 3 to 6 inputs and 1 or 2 outputs. The limited number of inputs is to avoid the exponential explosion. They are often implemented via SRAM. Each LUT can also have an associated flip-flop.

As such a field Programmable Gate Array is collection of a large number of **CLB**s (*Configurable Logic Blocks*, in AMD terminology) or **LE**s (*Logic Elements*, in Intel terminology). CLBs and LEs are smaller than a macrocell, organized as a matrix array of rows and columns. Programmable interconnections allow to route signals from one CLB to another, with wires running between rows and columns. Configurable I/O blocks allow to connect to the external world, in a similar fashion to CPLDs.

1.2.1 Logic Elements

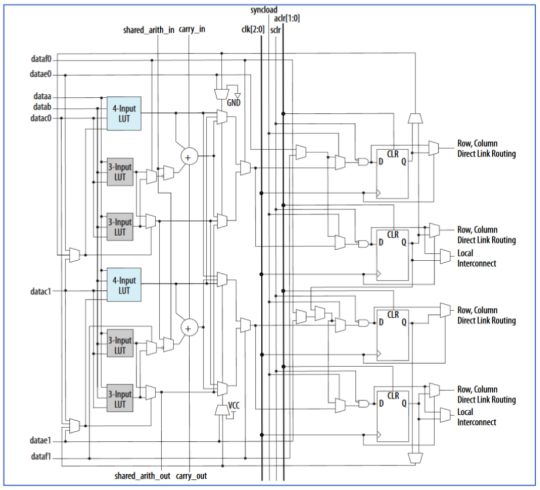
Let's stick to Intel terminology. A **logic element** is a device which can have many different designs, with the figure shown being a simple example.

Usually, an LE comibnes a LUT with output selection circuitry (a multiplexer, which might choose between standard outputs and synchronized ouputs). The multiplexer configuration is usually based on the same technology as LUTs (SRAM in most cases).

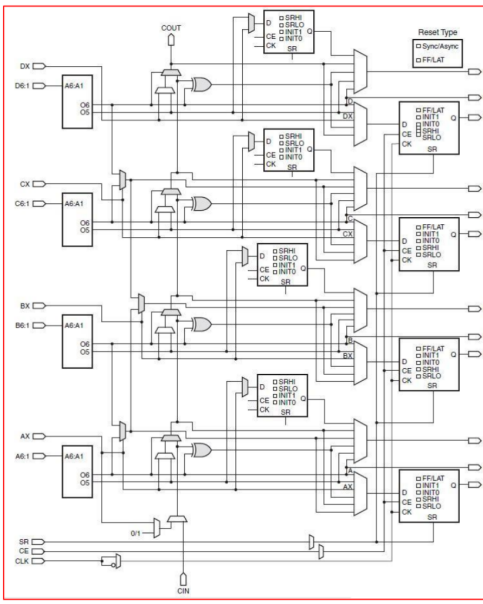


Field Programmable Gate Array (FPGA)

- Logic Element
  - Advanced LEs and CLBs from Intel/Altera and AMD/Xilinx



Intel/Altera ALM



AMD/Xilinx CLB